# Zhaofu (Jeff) Chen

Materials research | Conducting polymers | ENZ optics

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## EDUCATION

**University of California, Irvine** — **Expected Jun 2027**
*B.S. Chemical & Biomolecular Engineering; Minor in Materials Science and Engineering* — *Irvine, CA*
*Henry Samueli School of Engineering*

## PUBLICATIONS & MANUSCRIPTS

*Peer-reviewed publications*

Guo, H., Wang, X., Xu, M., **Chen, Z.**, Du, C., Wardini, J. L., Lam, B., Luong, J., Aoki, T., Pan, X., and Bowman, W. J. **3D nanoscale design of oxide morphology, anisotropy and chemical heterogeneity by glancing angle physical vapor deposition.** *Nanoscale* (2026). doi: 10.1039/d6nr01275b.

Guo, L., **Chen, Z.**, Han, H., Liu, G., Luo, M., Cui, N., Dong, H., and Li, M.-Z. **Advances and outlook in modified graphene oxide (GO)/epoxy composites for mechanical applications.** *Applied Nanoscience* **13**, 3273–3287 (2023). doi: 10.1007/s13204-022-02653-w.

*Submitted manuscripts*

Anopchenko, A., Yan, J., Shim, H. J., Dang, Q. T., Gonzalez, C. M., **Chen, Z.**, Genc, S., Khodagholy, D., and Lee, H. W. H. **Ultrafast-Laser Control of Optical Nonlinearities through De-Doping in Conducting Polymers.** *Submitted to Nature Materials.*

Ziff, A.†, Lin, L.†, **Chen, Z.**†, Shafer, H.†, Guo, H., and Bowman, W. J. **Solid-state synthesis & characterization of indium-oxide-based medium-entropy oxides.** ***Co-first author, equal contribution†.*** *Manuscript submitted.*

*Manuscripts in preparation*

**Tunable Near-Infrared Epsilon-Near-Zero Response in PEDOT:PSS Films via Thermal and Ethylene-Glycol Processing.** ***Co-first author, equal contribution.*** *In preparation for submission to Advanced Optical Materials; working title.*

**Tunable Conducting Polymers for Nonlinear and Epsilon-Near-Zero Optics: A Review.** *In preparation for submission to Laser & Photonics Reviews.*

**Thermal-Stability Design Strategies for Non-Fullerene Organic Photovoltaics.** *Review manuscript in preparation; working title.*

**Amorphous high-entropy oxides for $CO_2$ electroreduction.** ***Co-first author, equal contribution.*** *In preparation; planned submission to Advanced Materials.*

## CONFERENCES & PRESENTATIONS

- **APS Far West Section Meeting** — 2026
  Presenting author; poster contribution. **Chen, Z.**, Yan, J., Anopchenko, A., Chao, A., and Lee, H. W. "Tailoring Near-Infrared Epsilon-Near-Zero Optical Resonances in Conducting Polymer Thin Films via Controlled Processing." [Poster PDF]
- **UCI UROP Symposium & IRT Project Symposium** — Jun 2025; Jun 2026
  Research presentations at both symposia in 2025 and 2026. [2026 IRT Poster PDF; co-author]
- **UCI Engineering Council Research Fair** — Nov 2025
  Poster presentation.
- **Optica Advanced Photonics Congress** — Co-author on conference submission.
- **SPIE Nanoscience + Engineering** — Poster co-author: "Enhanced Third-Harmonic Generation in PEDOT:PSS via Femtosecond-Laser-Induced Reduction." [Poster PDF]

## RESEARCH EXPERIENCE

### Lee Nano-Optics Lab — **2025 – Present**

*Undergraduate Researcher · Advisor: Prof. Ho Wai (Howard) Lee* — *University of California, Irvine*

- ***Tunable Near-Infrared ENZ in PEDOT:PSS Thin Films***
  - Fabricate PEDOT:PSS films and vary annealing and ethylene-glycol treatment to tune their optical response.
  - Analyze ellipsometry data to extract film thickness, optical constants, and ENZ wavelengths; **co-first author** on the manuscript in preparation.

- ***Femtosecond Laser Writing & Third-Harmonic Generation in PEDOT:PSS***
  - Prepare PEDOT:PSS films and measure laser-induced third-harmonic generation using nonlinear optical microscopy.
  - Measure film thickness, optical constants, and conductivity for the manuscript submitted to *Nature Materials*. [SPIE Poster PDF; co-author]
- ***Nanoscale Transport in PEDOT:PSS via AFM/c-AFM***
  - Prepare films on conductive substrates for AFM/c-AFM measurements of surface morphology and local current.
  - Analyze AFM maps and step-height profiles in Gwyddion to compare film thickness and laser-modified regions.
- ***ITO/AZO ENZ Thin Films for Mid-IR Sampling*** *2026 – Present*
  - Anneal ITO and fabricate AZO by ALD for mid-infrared ENZ experiments.
  - Use ellipsometry to determine optical constants and ENZ wavelengths for a UC Irvine–Max Planck Institute collaboration.
- ***Conducting-Polymer Review for Nonlinear & ENZ Optics***
  - Review conducting-polymer nonlinear and ENZ optics; prepare figures and revise manuscript sections.

### Nguyen Lab — Summer 2026

*Summer Researcher · Advisor: Prof. Thuc-Quyen Nguyen* — *University of California, Santa Barbara*

- ***Crosslinking & Thermal Stability in Organic Photovoltaics***
  - Compared crosslinking chemistries, processing conditions, and device-aging results from the literature; contributed to a review manuscript.
  - Proposed experiments to test how crosslinker loading and activation affect active-layer morphology, charge transport, and thermal stability.

### Bowman Lab — 2024 – Present

*Undergraduate Researcher · Advisor: Prof. William Bowman* — *University of California, Irvine*

- ***Indium-Oxide-Based Medium-Entropy Oxides***
  - Analyze XRD data to identify phases and compare sintering conditions; develop Python/MATLAB plotting workflows.
  - **Co-first author** on the submitted manuscript.
- ***Amorphous High-Entropy Oxides for $CO_2$ Electroreduction***
  - Fit EIS data and analyze XRD, XPS, and SEM–EDS results; develop four-probe measurement workflows.
  - Lead proposal development and contribute to manuscript preparation as a **co-first author**. [CCAM Poster PDF; co-author]
- ***Complex-Concentrated-Oxide Memristors*** *2025; completed*
  - Led proposal development and experimental planning for oxide-based memristor devices.
  - Automated I–V analysis, switching-cycle statistics, and resistance-state comparisons.
- ***YSZ Thin Films & Glancing-Angle Deposition Architectures***
  - Analyzed XRD data and automated texture comparisons for YSZ films; contributed to the *Nanoscale* publication.

## INDUSTRY EXPERIENCE

**GoldenKeys High-tech Materials Co., Ltd.** — **Jul – Aug 2024**

*Technical Control & Quality Intern*

- Automated Opentrons FLEX testing and MATLAB/Python data processing for oxidation-catalyst development; implemented data schemas and quality checks to improve traceability.

## TECHNICAL SKILLS

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|---|---|
| **Thin-film fabrication** | PLD; ALD AZO; ITO annealing; PEDOT:PSS formulation; spin/drop coating; EG/MLLC processing; GOPS crosslinking; femtosecond laser writing; acid treatment. |
| **Characterization** | THG microscopy (Leica SP8); spectroscopic ellipsometry; Raman; optical transmission/absorption; XRD; XPS; SEM–EDS; AFM/CAFM; TEM/STEM-HAADF; four-probe and I–V measurements; EIS. |
| **Analysis & software** | Optical-constant extraction; EIS fitting; microscopy image analysis; regression and curve fitting; Python; MATLAB; Origin; ImageJ; Gwyddion; Adobe Illustrator; Excel. |

## HONORS, FELLOWSHIPS & RESEARCH GRANTS

- **Summer QUROP Award**, Eddleman Quantum Institute, UCI — 2026 | $3,000
- **MSA Undergraduate Research Scholarship** — 2026 | $3,000
- **CALIT2 UROP IRT Research Grant**, Project Lead — 2025–2026 | $3,000
- **CALIT2 IRT Research Grant** — 2024–2025 | $3,000

*UROP Researcher (2024–2025); UROP Project Lead (2025); CALIT2 IRT Research Fellow (2024–Present).*